

Unit-4 Sequential circuit

diff. b/w combinational and sequential ckt.

combinational

- o/p at any instant of time are dependent on present i/p.
- It doesn't require memory element.
- It doesn't require clock pulse.
- It may form a part of sequential of ckt.

sequential

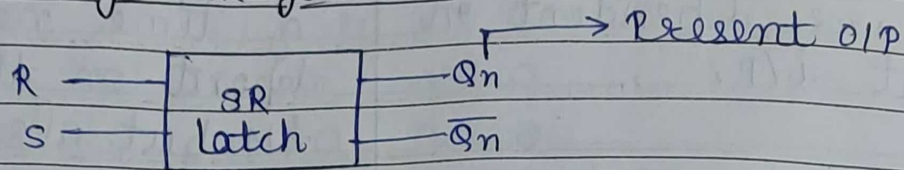
- o/p at any instant of time not only depend on the present i/p but also on the past i/p, o/p.
- Always contain memory elements.
- It synchronous sequential ckt also required clock pulses but asynchronous sequential ckt don't require clock pulses.
- But reverse is not true.

Latch v/s flip-flop :-

Latch is a 1 bit storage device
Spiral

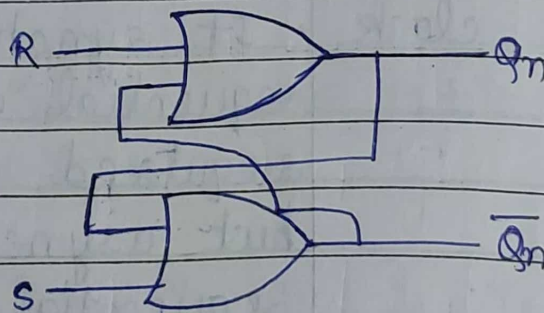
It can store 1 bit (0,1) information. Latch is a bi-stable ckt which responds to change of i/p. Logic levels as they occur, where as flip-flop only responds to i/p when they are enable by clock pulse.

SR latch using NOR gate ÷



Block diagram

ckt diagram ÷



A	B	NOR
0	0	1
0	1	0
1	0	0
1	1	0

$S = \text{Set} = 1, Q_n = 1$

$R = \text{Reset} = 1, Q_n = 0$

Truth table ÷

S	R	Q_{n+1}	$\overline{Q_{n+1}}$
0	0	Q_n	$\overline{Q_n}$
0	1	0	1
1	0	1	0
1	1	d	d

Characteristics table ÷

	S	R	Q_n	Q_{n+1}	
0	0	0	0	0	] memory
1	0	0	1	1	
2	0	1	0	0	] Reset
3	0	1	1	0	
4	1	0	0	0	] set
5	1	0	1	1	
6	1	1	0	d	] forbidden state
7	1	1	1	d	

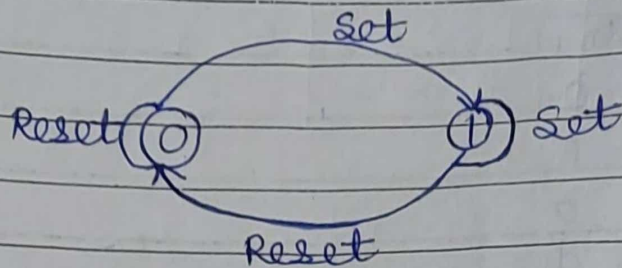
$Q_n = \text{Present o/p}$, $Q_{n+1} = \text{Next o/p state}$

characteristic equation ÷

S \ R	00	01	11	10
0		1		
1	1	1	d	d

$$Q_{n+1} = S + \overline{R}Q_n$$

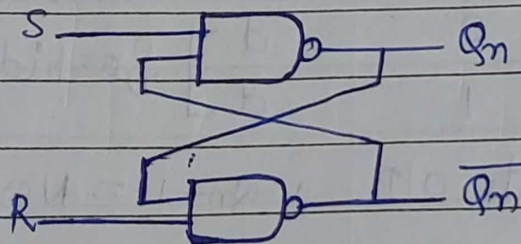
State diagram :-



Excitation table :-

Q_n	Q_{n+1}	S	R
0	0	0	d
0	1	1	0
1	0	0	1
1	1	d	0

SR latch using NAND gate :-



Truth table :-

i/P		o/P		A	B	NAND
S	R	Q_n	Q_{n+1}			
0	0	d	d	0	0	1
0	1	1	0	0	1	1
1	0	0	1	1	0	1
1	1	Q_n	Q_n	1	1	0

Characteristic table ÷

	S	R	Q_n	Q_{n+1}	
0	0	0	0	0	] don't care
1	0	0	1	1	
2	0	1	0	0	] set
3	0	1	1	1	
4	1	0	0	0	] reset
5	1	0	1	1	
6	1	1	0	0	] memory
7	1	1	1	1	

Excitation table ÷

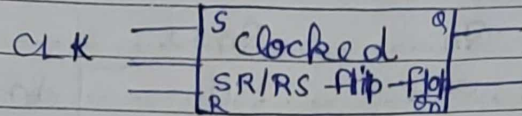
Q	Q_{n+1}	S	R
0	0	1	0
0	1	0	1
1	0	1	0
1	1	0	0

Characteristic equation ÷

S \ R	Q_n 00	01	11	10
0	0	1	0	1
1	1	0	1	0

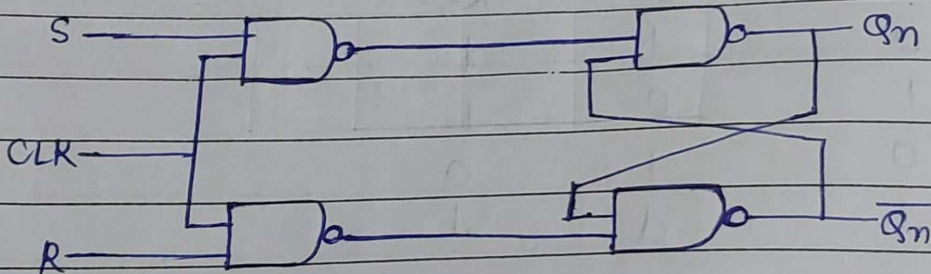
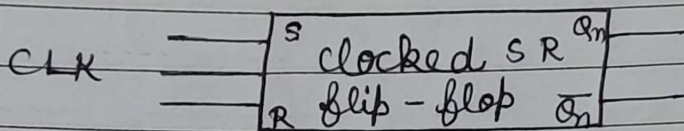
$$Q_{n+1} = \bar{S} + RQ_n$$

Clocked SR/RS flip-flop :-



It RS or SR latch is modified with additional clock input (i.e. clock pulses) that controls the change of state in a flip-flop. This modified RS/SR flip-flop is also called as clocked SR/RS flip-flop.

It can be implemented using NAND or NOR gate.

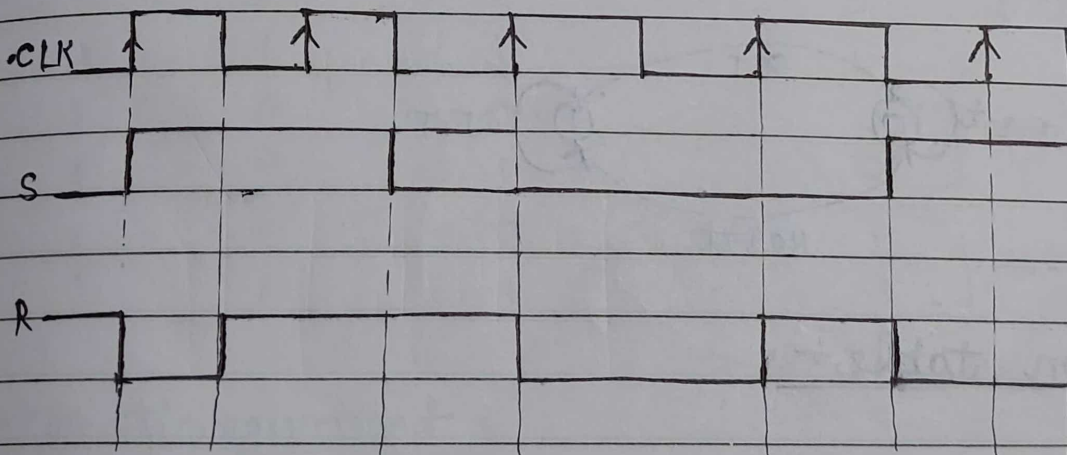


Truth table :-

I/P			O/P		
CP	S	R	Q_{n+1}	$\overline{Q_{n+1}}$	
0	X	X	Q_n	$\overline{Q_n}$	No change memory
1	0	0	Q_n	$\overline{Q_n}$	
1	0	1	0	1	Reset
1	1	0	1	0	Set
1	1	1	d	d	Forbidden state

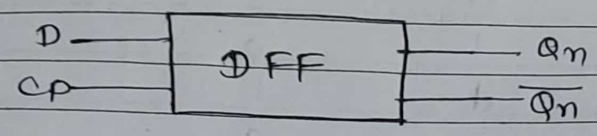
characteristic table :-

	CLK	S	R	$Q_m (PS)$	Q_{m+1}	
0	0	0	0	0	1	No change memory
1	0	0	0	1	1	
2	0	0	1	0	0	
3	0	0	1	1	1	
4	0	1	0	0	0	
5	0	1	0	1	1	
6	0	1	1	0	0	
7	1	0	0	1	1	
8	1	0	0	0	0	
9	1	0	1	0	0	Reset
10	1	0	1	1	0	
11	1	1	0	0	1	set
12	1	1	0	1	1	
13	1	1	1	0	d	don't care
14	1	1	1	1	d	
15	1	1	1	1	1	



D flip-flop :-

To avoid the forbidden state in the SR flip-flop, we can use a D flip-flop. It is also called as display or data flip-flop. It is delay the I/P for a certain time period depending upon the clock triggering, before transmitting the I/P to the O/P. Therefore, termed as delay flip-flop.

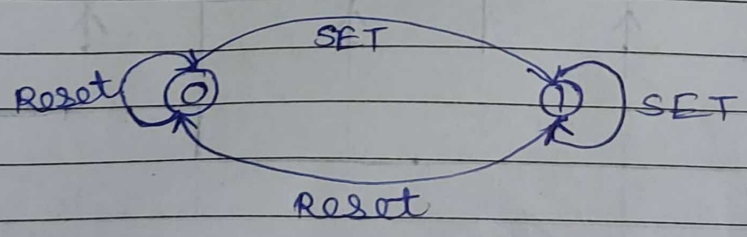


characteristics eqⁿ :-

for $C=1$

S \ R	$Q_n = 0$	$Q_n = 1$	$Q_n = 1$	$Q_n = 0$
0	0	1	d	d
1	1	1	d	d

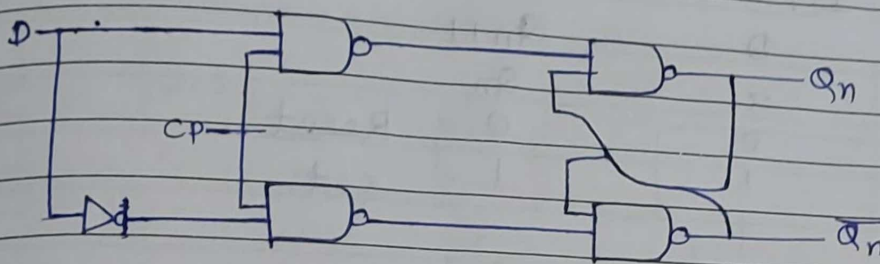
$$Q_{n+1} = S + \bar{R} Q_n$$



excitation table :-

(PS)	$Q_{n+1}(NS)$	S	R
0	0	0	0
0	1	0	1
1	0	1	0
1	1	1	1

waveform ÷



characteristics eqⁿ. ÷

	CLK	D	Q_n (PS)	Q_{n+1} (MS)	
0	0	0	0	0	No change
1	0	0	1	1	
2	0	1	0	0	
3	0	1	1	1	
4	1	0	0	0	Reset
5	1	0	1	0	
6	1	1	0	1	Set
7	1	1	1	1	

characteristic equation ÷

$D \backslash Q_n$	0	1
0	0	0
1	1	1

$$Q_{n+1} = D$$

Truth table ÷

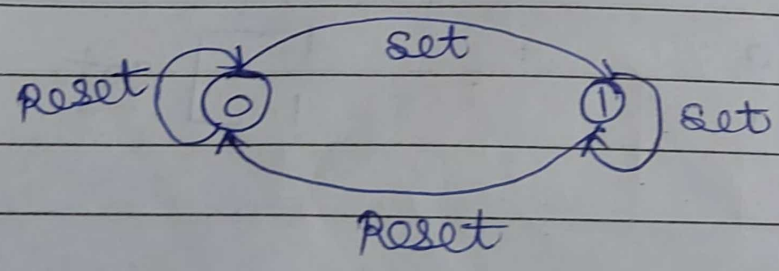
I/P		O/P
C_p	D	Q_{n+1}
0	0	0
1	0	0
1	1	1

Reset
set

Excitation table ÷

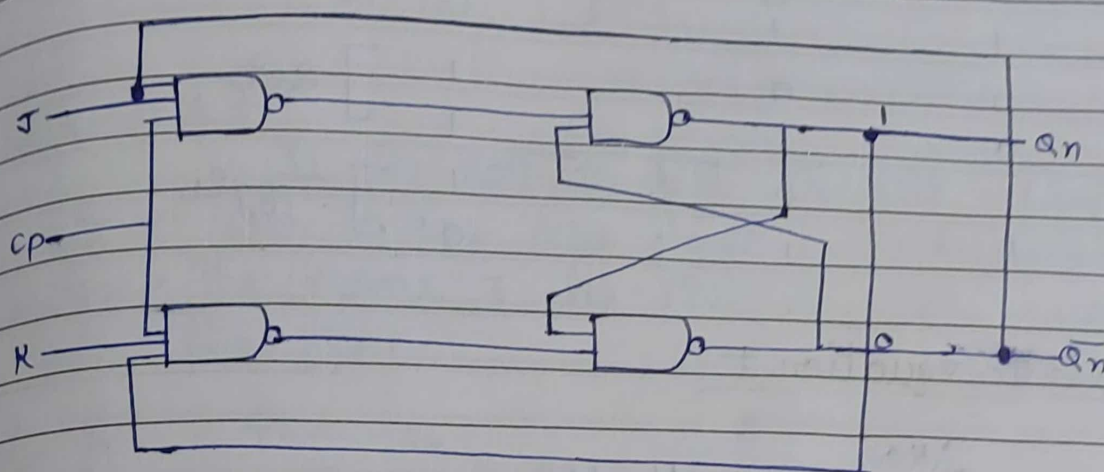
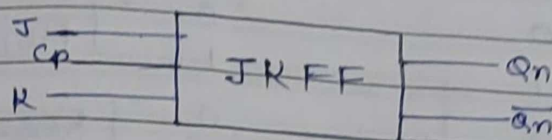
$Q_n (PS)$	$Q_{n+1} (MS)$	D
0	0	0
0	1	1
1	0	0
1	1	1

State diagram ÷



JK flip-flop :-

It is modified form of SR flip-flop. It has similar characteristics as that of SR flip-flop except that intermediate or invalid state.



Truth table :-

CP	J	K	Q_{n+1}	Mode
0	X	X	Q_n	No change
1	0	0	Q_n	
1	0	1	0	→ Reset
1	1	0	1	→ Set
1	1	1	$\overline{Q_n}$	→ Toggle

characteristics table :-

Assume $C = 1$

	J	K	Q_n (PS)	Q_{n+1} (NI)	
0	0	0	0	0	
1	0	0	1	1	] Memory
2	0	1	0	0	
3	0	1	1	0	] Reset
4	1	0	0	1	
5	1	0	1	1	] Set
6	1	1	0	1	
7	1	1	1	0	] Toggle

Characteristic equation :-

J \ K	00	01	11	10
0		1		
1	1	1		1

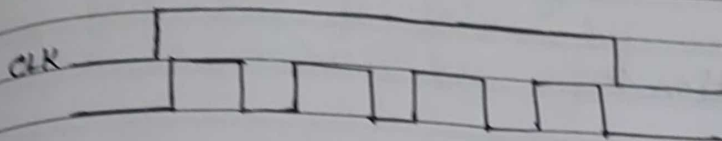
$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

Excitation table :-

Q_n (PS)	Q_{n+1} (MS)	J	K
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0

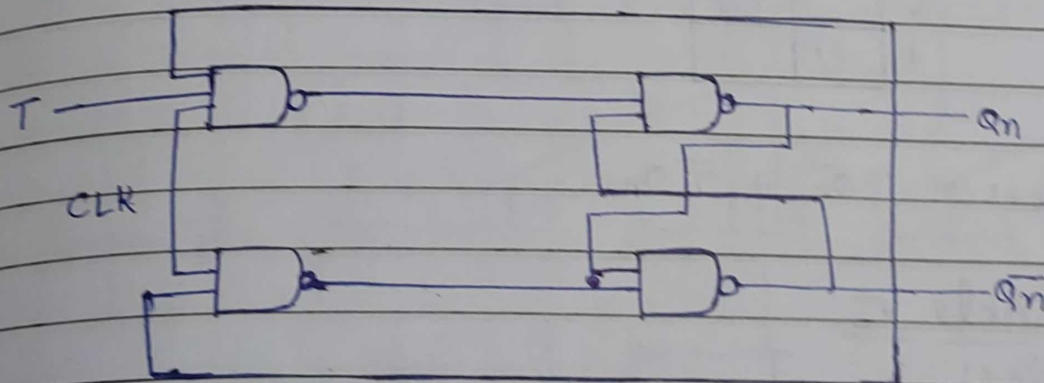
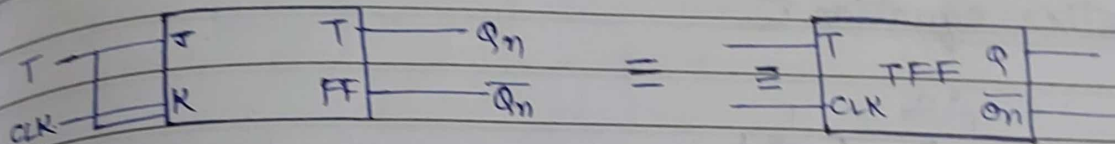
Race around condition +

If $J = K = 1$, and clock = 1 for a long period of time then the o/p Q will toggle as long as clock remains high which makes the o/p unstable or uncertain. This is called race around condⁿ.



T flip-flop +

It is nothing but JK flip-flop where both i/p J and K are tied together and single i/p is assigned. The name T flip-flop is also known as toggle flip-flop.



Truth table :-

CLK	T	Q_{n+1}	Note
0	X	Q_n	No change
1	0	Q_n	No change
1	1	$\overline{Q_n}$	Toggle

Characteristics equation :-

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Characteristics equation :-

T	Q_n	0	1
0			1
1		1	

$$Q_{n+1} = T\overline{Q_n} + \overline{T}Q_n$$

Excitation table :-

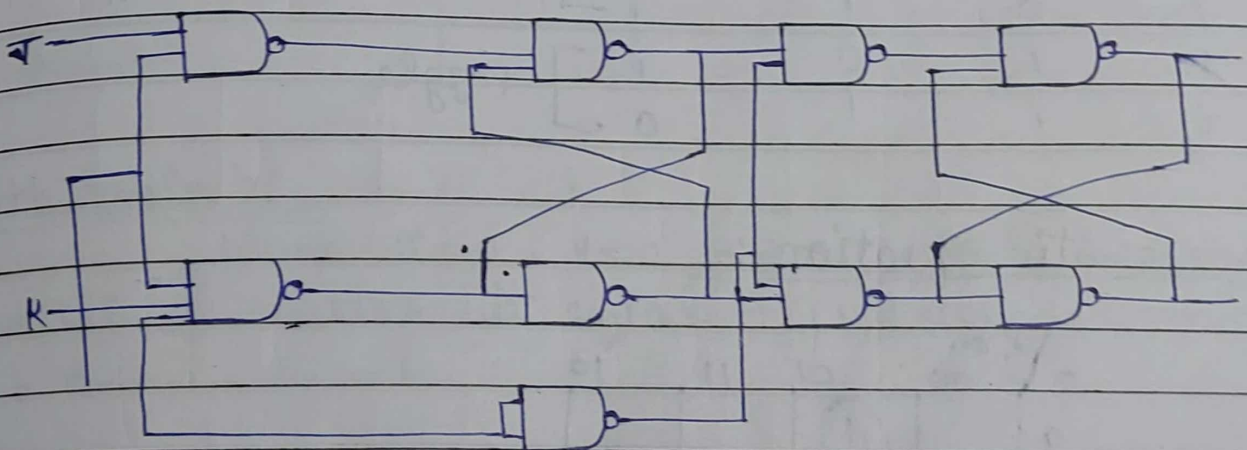
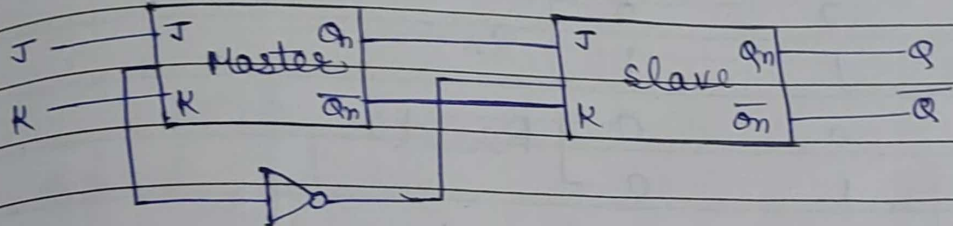
Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Master-slave JK flip-flop:

It is constructed by 2 JK flip-flop. One FF work as master flip-flop and 2nd flip-flop work as slave flip-flop.

Master flip-flop is driven by (+ve) clock while slave FF is driven by (-ve) clock pulse.

The race around condⁿ. is avoided the feedback from slave to master & slave being inact during (+ve) half cycle.



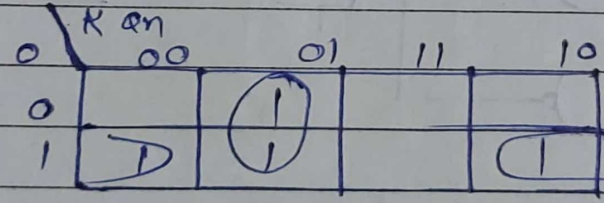
Truth table ÷

CLK	J	K	Q_{n+1}	
0	X	X	Q_n	(No change)
1	0	0	Q_n	(No change)
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	$\overline{Q_n}$	Toggle

Characteristic equation ÷ for $c = 1$

	J	K	Q_n	Q_{n+1}	
0	0	0	0	0	] Q_n
1	0	0	1	1	
2	0	1	0	0	] Reset
3	0	1	1	0	
4	1	0	0	1	] Set
5	1	0	1	1	
6	1	1	0	1	] Toggle
7	1	1	1	0	

Characteristic equation ÷

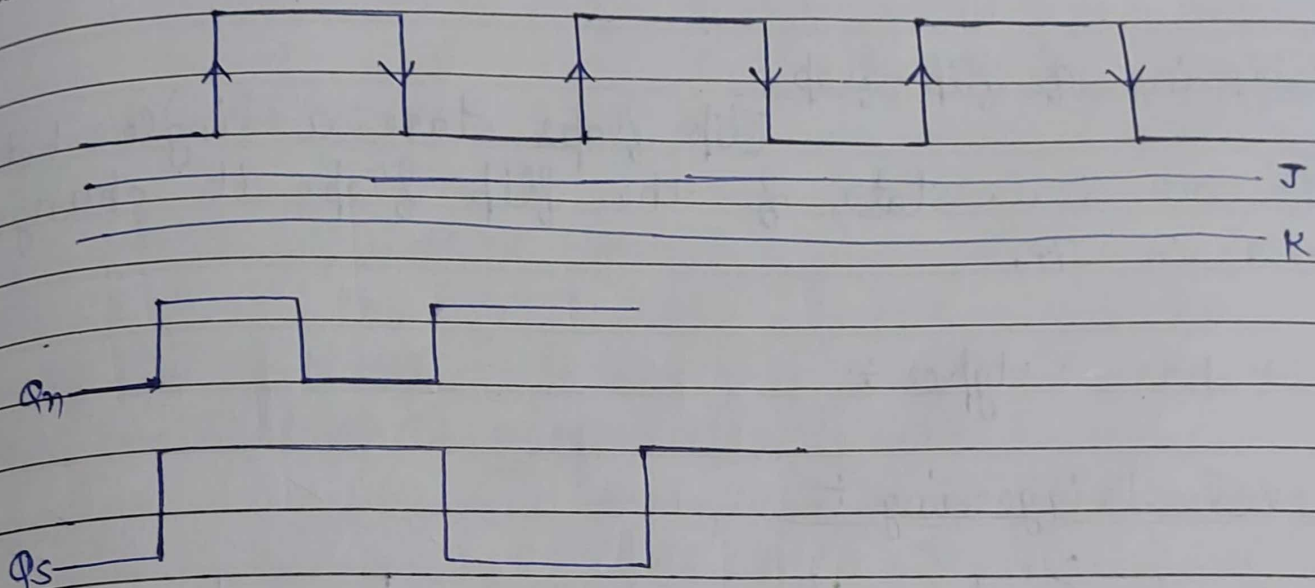


$$Q_{n+1} = J\overline{Q_n} + \overline{K}Q_n$$

Excitation table :-

Q	Q_{n+1}	J	$K \rightarrow$ i/p
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0

waveform :-



Truth table :-

When $J=0$, $K=0$ when clock is high, the o/p of master is equal to unchanged, that is 0, $Q_n=0$, $\bar{Q}_n=1$

o/p of the slave = un effective.

when clock is equal to low, o/p of the master = unaffected.

O/P of the slave = $Q_s = 0$ or $\bar{Q}_s = 1$
 $\downarrow$
 unchanged

$J=1, K=0$
 clock = high [O/P of master = set, $(Q_m=1, \bar{Q}_m=0)$
 [O/P of slave = unaffected]

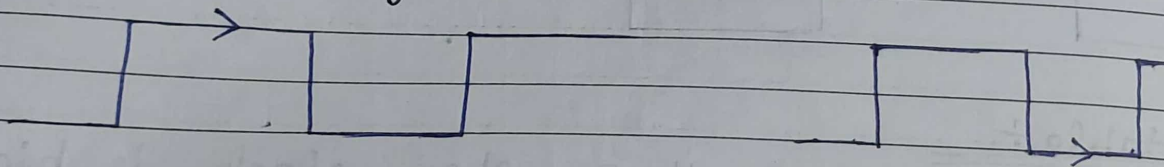
clock = low [O/P of master = unaffected
 [O/P of slave set]

Triggering of flip-flop:

The O/P i.e. state of the flip flops it change.
 Flip flops store a single bits.

It has 2 types:

(i) Level triggering:



(+)ve level triggering

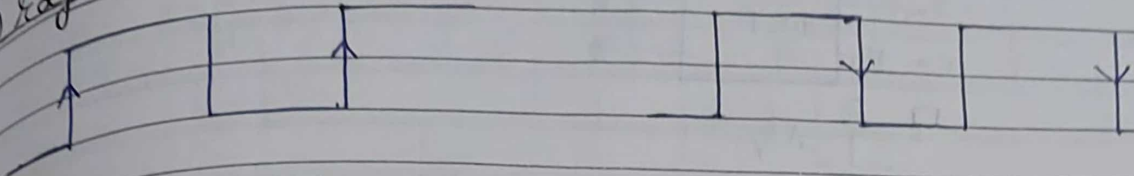
(-)ve level triggering

If the clocked flip-flop changes its state when clock is +ve that is high level is called (+)ve level triggering.

When clock is -ve or low level is called (-)ve level triggering.

The limitation of the level triggering is that as long as clock at level +ve or -ve, the flip flop will change its state as many times as i/p change. if i/p frequency is more than clocked frequency.

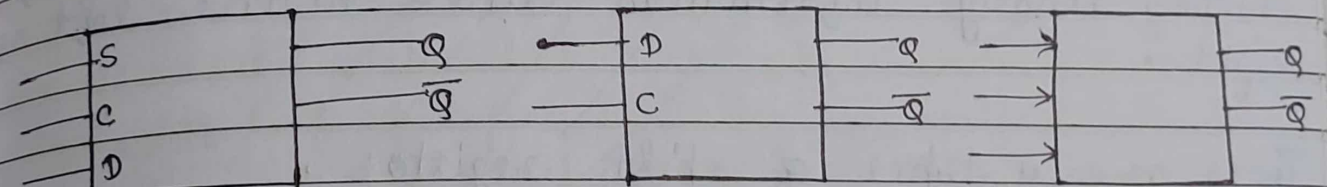
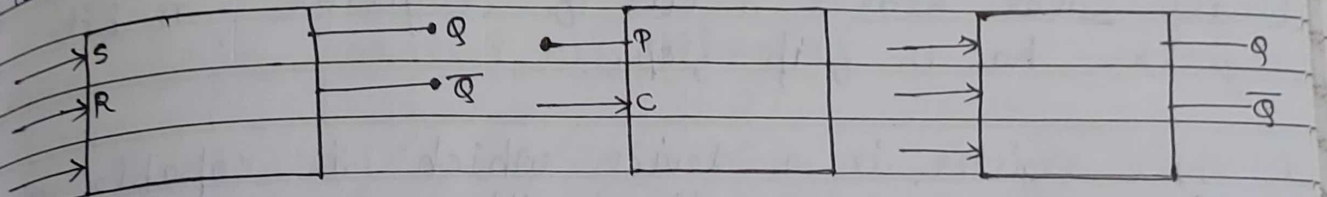
(2) Edge triggering :-



+ve edge triggering

-ve edge triggering

When clocked flip-flop change its o/p only, when clock make a transition from low to high to low is called A triggering.



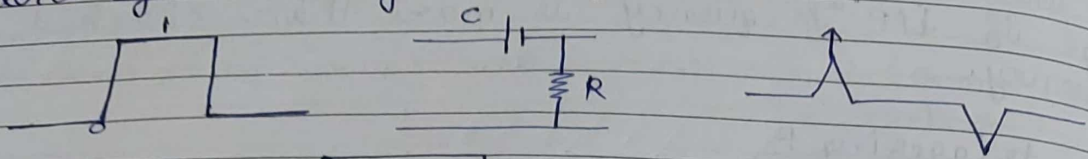
(a) S-R FF

(b) D-FF

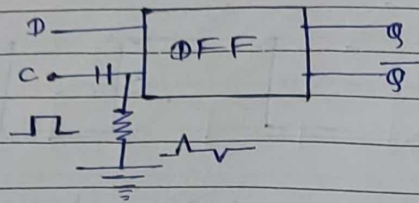
(c) J-K FF

Triggering of flip-flop :-

edge triggering can be done by inserting RC differentiator.



Ex:-



Shift register :-

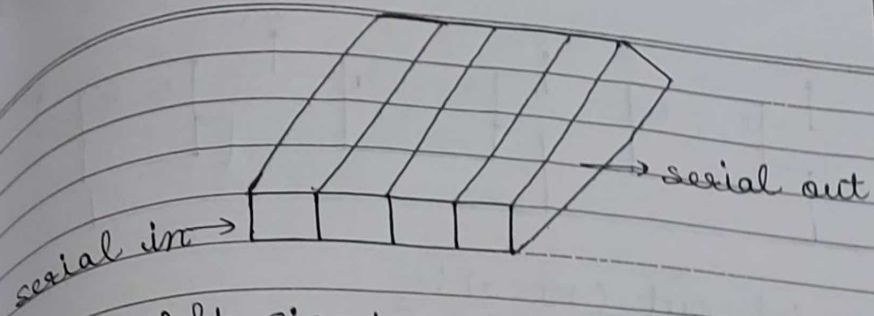
Register :-

It is a group of flip-flop or binary sales which holds the binary information. Since binary sales stores a bits of register, n bit register has n flip-flop.

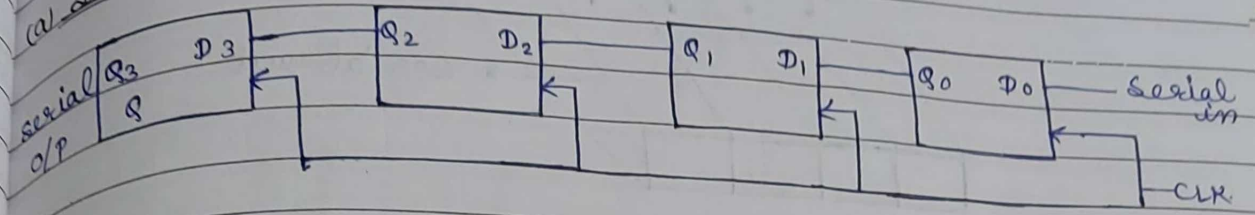
A shift register is a device which is capable of shifting binary information either to the left or right.

These are 4 types of shift register.

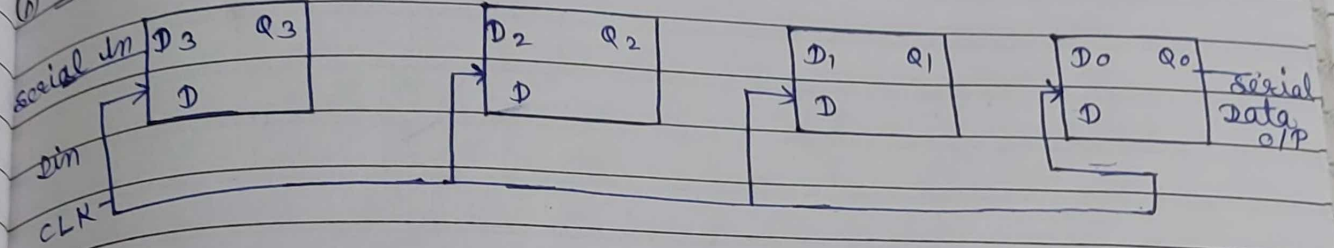
(i) Serial IN register serial out register (SISO) :-



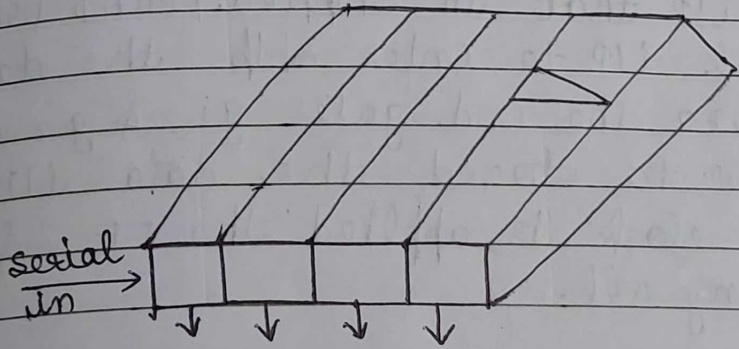
(a) Shift left SISO :

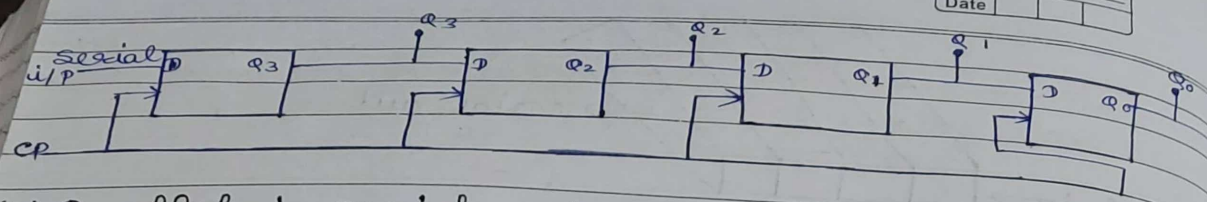


(b) Shift right SISO :

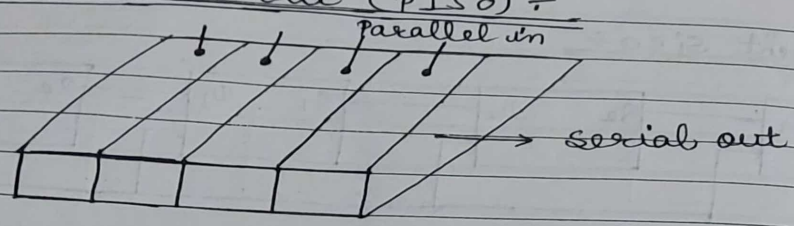


(c) Serial in parallel out (SIPO) :





(3) Parallel in serial out (PISO) :-



It has a control i/p that is Shift/LOAD which controls the parallel i/p to enter shift the data. When Shift/LOAD is zero the end gates g_1 & g_2 & g_3 gates enables & allow to stored the data i/p respective FF. When clock is applied the FF & then $D=0$ these by storing all.